

WMDR2

Package in package 'Model'

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WMDR2 diagram

Class diagram in package 'WMDR2'

WMDR2 v0.3.x

v0.3.0

- Cardinality of facility.deployments changed to [0..*]
- Deployment.location [0..1] added to describe relative location of instrument as free
- Deployment.serialNumber moved to ObservingConfiguration.InstrumentInstance.
- Deployment.instrument associated to InstrumentInstance
- Deployment.sourceOfObservation moved to ObservingConfiguration
- As a consequence of the above, Deployment describes the location context only and
- "reporting" folded into "ReportingProcedure"
- ReportingProcedure.validFrom dropped.
- ReportingProcedure.spatialReportingInterval added.
- cardinality contact.organization from [0..1] to 1
- cardinality wmoRegion from [0..1] to 1
- cardinality of ReportingProcedure.dataPolicy from [0..1] to 1; nilReason allowed
- cardinality of ReportingProcedure.internationalExchange from [0..1] to 1
- cardinality of ObservingConfiguration.operatingStatus from [0..1] to 1
- cardinality of ObservingConfiguration.sourceOfObservation from [0..1] to 1

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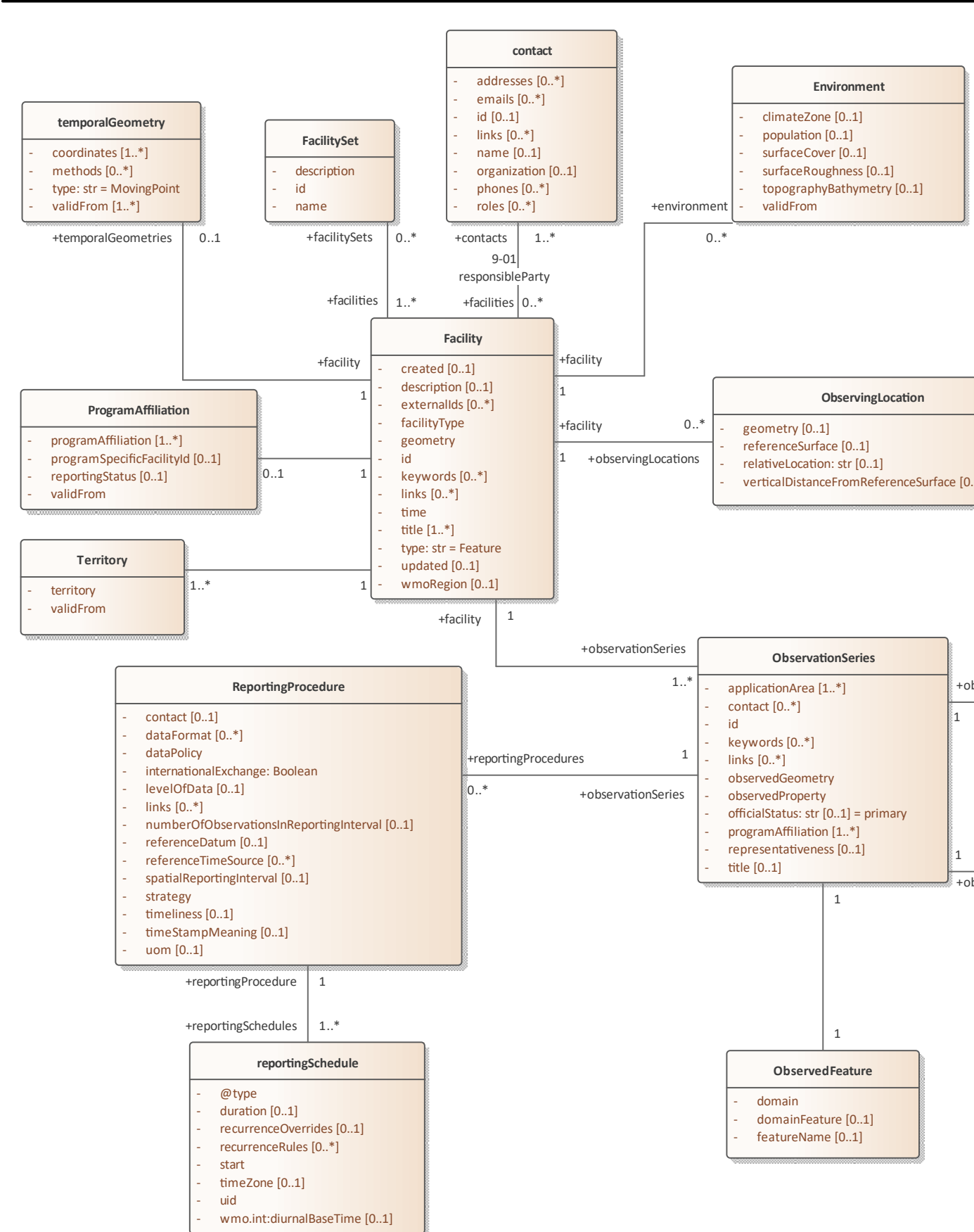


Figure 1: WMDR2

ATTRIBUTES

Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Email contacts.

[Is static

 id : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Optional stable contact identifier.

[Is static

 links : Private

Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Online resources associated with the contact.

[Is static

 name : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Either a full name, or a sequence of name parts. Examples:

Dr. Jörg Klausen (this corresponds to FN in vcard)

Klausen;Jörg;Dr.; (this corresponds to N in vcard)

[Is static

 organization : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Example:

ORG:Federal Office of Meteorology and Climatology MeteoSwiss

[Is static

 phones : Private

Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Environment

Class in package 'WMDR2'

Facility-environment object containing climate, surface-cover, population, and topo

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ATTRIBUTES		
	climateZone : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static
	population : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Population in 10 and 50 km radius around the facility	[Is static
	surfaceCover : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static
	surfaceRoughness : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static
	topographyBathymetry : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static
	validFrom : Private	

ASSOCIATIONS		
 Association (direction: Unspecified)		
Source: Public describes (Class) ObservationSeries		Target: Public
Cardinality: [1..*]		Cardinality
		ObservingProc

Facility

Class in package 'WMDR2'

Facility properties payload of the WMDR2 full record. Root Features geometry/time here.

ATTRIBUTES	
 created : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	
Record creation date/time metadata.	[Is static
 description : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	
Facility description.	[Is static

ATTRIBUTES	
geometry = geolocation = coordinates + elevation	[Is static]
 id : Private Alias: 3-06 (primary) WSI to prefixed with name space "facility:"	[Is static]
 keywords : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Discovery/search keywords retained in the core JSON.	[Is static]
 links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Related links.	[Is static]
 time : Private Alias: [dateEstablished, dateClosed mandatory element in OGC Records API Part 1, an interval. Used to declare [dateEstablished]	[Is static]
 title : Private Alias: 3-03 facilityName Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False) properties.title = 3-03 facilityName	[Is static]
 type : str. Private = Feature	

ASSOCIATIONS		
 Association (direction: Unspecified)		
Source: Public facilities (Class) Facility Cardinality: [1..*]		Target: Public Cardinality
 Association (direction: Unspecified)		
Source: Public facility (Class) Facility Cardinality: [1]		Target: Public ObservingLoc Cardinality
 Association (direction: Unspecified) 9-01 responsibleParty		
Source: Public facilities (Class) Facility Cardinality: [0..*]		Target: Public Cardinality
 Association (direction: Unspecified)		
Source: Public facility (Class) Facility Cardinality: [1]		Target: Public temporalGeom Cardinality
 Association (direction: Unspecified)		
Source: Public facility (Class) Facility Cardinality: [1]		Target: Public Cardinality
 Association (direction: Unspecified)		
Source: Public (Class) Facility Cardinality: [1]		Target: Public Cardinality
 Association (direction: Unspecified)		

ATTRIBUTES		
 description : Private		[Is static
 id : Private		[Is static
 name : Private Alias: title		[Is static
ASSOCIATIONS		
 Association (direction: Unspecified)		
Source: Public facilities (Class) Facility Cardinality: [1..*]		Target: Public Cardinality

instrument

Class in package 'WMDR2'

Describes instrument capabilities, not a specific instrument instance.

For semantic consistency, terms like observedProperty and observedGeometry are used for the instrument capabilities, not the actual observations. With this, instrument becomes a class and even facilities.

ATTRIBUTES	
 verticalRange : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	
Specify using an array of [lower boundary, upper boundary] in meter. This is a capability actually accessible vertical range.	
[Is static	

ASSOCIATIONS	
 Association (direction: Unspecified)	
Source: Public groups (Class) instrument Cardinality: [0..1]	Target: Public Cardinality
 Association (direction: Unspecified) 5-09 Instrument	
Source: Public instrumentInstances (Class) InstrumentInstance Cardinality: [0..*]	Target: Public Cardinality

InstrumentInstance

Class in package 'WMDR2'

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ATTRIBUTES	
 serialNumber : Private	
[Is static	

ASSOCIATIONS	
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ATTRIBUTES		
 description :	Private	[Is static
 id :	Private	[Is static
 name :	Private Alias: title	[Is static
ASSOCIATIONS		
	Association (direction: Unspecified)	
Source: Public groups (Class) instrument	Cardinality: [0..1]	Target: Public Cardinality:

ObservationSeries

Class in package 'WMDR2'

WMDR2 core JSON class: observation.

ATTRIBUTES		

ATTRIBUTES

 keywords : Private
 Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Observation keywords.

[Is static]

 links : Private
 Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

Observation links.

[Is static]

 observedGeometry : Private

cf. <http://codes.wmo.int/wmdr/Geometry>

[Is static]

 observedProperty : Private
 Alias: OMS:ObservedProperty, iop:Variable

The observedVariable is constructed following I-ADOPT ontology approaches, with elements vocabulary hosted at codes.wmo.int/wmdr2.

NB: observedVariable will be a LONG list of entries, each record comprised of notation, unique identifier, not necessarily human-readable. The name is potentially human-readable definition, human-readable and describing all relevant aspects of an observed variable. It observedProperty, the object of interest, the matrix and domain, as well as constraints. Many controlled vocabularies that may be domain-specific.

Examples include:

- observableProperty: state variable (e.g., temperature, pressure, leaf-area-index (LAI), ... thickness, concentration, scattering coefficient, color, ...)
- objectOfInterest: sea ice, CO2, aerosol particles, chlorophyll
- observablePropertyConstraints: a conventional level above ground (e.g., 2m above ground size fraction (e.g., PM2.5), a specific wavelength (e.g., 635 nm), a pressure (range), a temperature etc., as relevant for the result of the observation. Constraints shall be documented as name come from a code list, and the values shall come from a code list depending on the name,
- matrix: atmosphere, interstitial air in the root space, air in an ice core, sea surface, soil, ...

These elements will be supported by code lists from codes.wmo.int/wmdr2.

ATTRIBUTES		
 representativeness : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) http://codes.wmo.int/wmdr/Representativeness [Is static		
 title : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Human-readable title derived from domain, geometry, observed variable, and optional lab [Is static		
ASSOCIATIONS		
 Association (direction: Unspecified) Source: Public describes (Class) ObservationSeries Cardinality: [1..*] Target: Public Cardinality: ObservingProc		
 Association (direction: Unspecified) Source: Public describes (Class) ObservationSeries Cardinality: [1] Target: Public Cardinality:		
 Association (direction: Unspecified) Source: Public observationSeries (Class) ObservationSeries Cardinality: [1] Target: Public ObservingProc Cardinality:		
 Association (direction: Unspecified)		

ASSOCIATIONS		
	Source: Public (Class) ObservationSeries Cardinality: [1]	Target: Public Cardinality
	describes	has_geolocalin
 Association (direction: Unspecified)		
	Source: Public groups (Class) ObservationSeries Cardinality: [1..*]	Target: Public ObservationSe Cardinality
 Association (direction: Unspecified)		
	Source: Public observationSeries (Class) ObservationSeries Cardinality: [1]	Target: Public ObservingCon Cardinality
 Association (direction: Unspecified)		
	Source: Public facility (Class) Facility Cardinality: [1]	Target: Public ObservationSe Cardinality

ObservationSeriesSet

Class in package 'WMDR2'

grouping of observations (which are themselves time series)

ATTRIBUTES

ASSOCIATIONS



Association (direction: Unspecified)

Source: Public groups (Class) ObservationSeries
Cardinality: [1..*]

Target: Public
ObservationSe
Cardinality

ObservedFeature

Class in package 'WMDR2'

Describes the physical domain, i.e. larger-scale environmental target or ultimate-fea

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ATTRIBUTES



domain : Private

The largest scale in WIGOS, cf. <http://codes.wmo.int/wmdr/Domain>

[Is static



domainFeature : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Describes the type of the larger-scale environmental target or ultimate-feature-of-interest.

Concept motivated by requirements from hydrology, but may be useful elsewhere. New c

- river basin
- lake
- river
- catchment

- canopy

- epilimnion

ASSOCIATIONS



Association (direction: Unspecified)

Source: Public (Class) ObservationSeries
Cardinality: [1]

Target: Public
Cardinality

ObservingConfiguration

Class in package 'WMDR2'

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ATTRIBUTES



exposure : Private

Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Values from <http://codes.wmo.int/wmdr/Exposure>

[Is static



observingMethod : Private

from (new) unified code list of observing methods, based on
<http://codes.wmo.int/wmdr/ObservingMethodAtmosphere>
<http://codes.wmo.int/wmdr/ObservingMethodTerrestrial>
<http://codes.wmo.int/wmdr/ObservingMethodEarth>

[Is static



operatingStatus : Private

values refer to <http://codes.wmo.int/wmdr/InstrumentOperatingStatus>

[Is static

ASSOCIATIONS		
 Association (direction: Unspecified)		
Source: Public instrumentInstance (Class) ObservingConfiguration Cardinality: [0..1]		Target: Public InstrumentInst Cardinality
 Association (direction: Unspecified)		
Source: Public observationSeries (Class) ObservationSeries Cardinality: [1]		Target: Public ObservingCon Cardinality

ObservingLocation

Class in package 'WMDR2'

Documents the placement of a particular instrument type (from a catalogue) at a specific `relativeLocation` for the purpose of making measurements. Can also describe the position

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ATTRIBUTES
 geometry : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)
geometry is the OGC Records name for geolocation.
For stationary facilities, this represents the geolocation of the sensor / inlet, and should be geometry (and temporalGeometry) of the facility is assumed. For mobile facilities, this should be [Is static

ATTRIBUTES

{"value": float, "uom": str}, whereby "uom" from <http://codes.wmo.int/wmdr/unit>
[Is static

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public observingConfigurations (Class)
ObservingConfiguration
Cardinality: [0..*]

Target: Public
ObservingLoc
Cardinality

 Association (direction: Unspecified)

Source: Public facility (Class) Facility
Cardinality: [1]

Target: Public
ObservingLoc
Cardinality

ObservingProcedure

Class in package 'WMDR2'

When are observations made? Describes an observingStrategy, and depending on it, "from Feb-Oct on working days from 06-18 UTC".

Implement as array of objects

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ATTRIBUTES

 strategy : Private

Values from <http://codes.wmo.int/wmdr/SamplingStrategy>

ASSOCIATIONS		
Source: Public observationSeries (Class) ObservationSeries		Target: Public ObservingProc
Cardinality: [1]		Cardinality

observingSchedule

Class in package 'WMDR2'

Reusable JSCalendar Event schedule used by deployment.temporalObservingSchedule extensions are represented with wmo.int:* properties.

May be expected for strategy "continuous" or "discrete"

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ATTRIBUTES		
 @type : Private		
JSCalendar object type; const Event.		[Is static
 duration : Private		
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)		
Occurrence duration describing coverage window or sampling duration.		[Is static
 recurrenceOverrides : Private		
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)		
JSCalendar recurrence override patches; null override values are invalid.		[Is static

ATTRIBUTES		
		[Is static
 uid : Private	JSCalendar-safe schedule uid.	[Is static
 wmo.int:aggregationInterval : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	Describes the smallest interval over which raw/ level 0 observations are aggregated and s Implement as {"value": int, "uom": str}	[Is static
 wmo.int:samplingFrequency : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	The frequency at which raw / level 0 observations are obtained. Implement as {"value": int, "uom": str}	[Is static
ASSOCIATIONS		
 Association (direction: Unspecified)	Source: Public observingProcedure (Class) ObservingProcedure Cardinality: [1]	Target: Public observingSche Cardinality

ProgramAffiliation

Class in package 'WMDR2'

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ATTRIBUTES	
 programAffiliation : Private Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)	[Is static
 programSpecificFacilityId : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static
 reportingStatus : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) http://codes.wmo.int/wmdr/ReportingStatus	[Is static
 validFrom : Private	[Is static

ASSOCIATIONS	
 Association (direction: Unspecified) Source: Public (Class) Facility Cardinality: [1]	Target: Public Cardinality:

reporting

Class in package 'WMDR2'

Reusable reporting template.

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ASSOCIATIONS		
	Association (direction: Unspecified) 7-03 Temporal reporting	
Source: Public procedure (Class) ReportingProcedure		Target: Public
Cardinality: [1]		Cardinality

ReportingProcedure

Class in package 'WMDR2'

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ATTRIBUTES	
 contact : Private	
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static
 dataFormat : Private	
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)	
Aligned array, values from http://codes.wmo.int/wmdr/DataFormat .	
	[Is static

ATTRIBUTES	
Value from http://codes.wmo.int/wmdr/LevelOfData .	[Is static
 links : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) URL to related WIS record, entry point for data discovery and retrieval.	[Is static
 numberOfObservationsInReportingInterval : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)	[Is static
 referenceDatum : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) Value from http://codes.wmo.int/wmdr/CoordinateReferenceSystem .	[Is static
 referenceTimeSource : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) Single value or array, values from http://codes.wmo.int/wmdr/ReferenceTime .	[Is static
 spatialReportingInterval : Private Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False) To be implemented as {"value": [float], "uom": str}, whereby len([float])<3 to report either 1- or 2- dimension.	[Is static
 strategy : Private	[Is static

ATTRIBUTES		
		[Is static
ASSOCIATIONS		
 Association (direction: Unspecified)		
Source: Public reportingProcedure (Class) ReportingProcedure Cardinality: [1]		Target: Public reportingSchedule Cardinality: [1]
 Association (direction: Unspecified) 7-03 Temporal reporting		
Source: Public procedure (Class) ReportingProcedure Cardinality: [1]		Target: Public reportingSchedule Cardinality: [1]
 Association (direction: Unspecified)		
Source: Public observationSeries (Class) ObservationSeries Cardinality: [1]		Target: Public reportingProcedure Cardinality: [1]

reportingSchedule

Class in package 'WMDR2'

Reusable JSCalendar Event schedule used by deployment.temporalObservingSchedule. Extensions are represented with wmo.int:* properties.

ATTRIBUTES

ATTRIBUTES

 recurrenceRules : Private
Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

JSCalendar recurrence rules.

To be used for "reportingInterval"; use start, duration and other elements to evaluate consistency.
[Is static]

 start : Private

Canonical JSCalendar recurrence anchor; date 0001-01-01 with time from coverage start.
[Is static]

 timeZone : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

IANA time zone or UTC default.

[Is static]

 uid : Private

JSCalendar-safe schedule uid.

[Is static]

 wmo.int:diurnalBaseTime : Private
Multiplicity: ([0..1], Allow duplicates: 0, Is ordered: False)

Relevant for 24h aggregates, e.g., precipitation amount collected over 24h.

Implement as {"time": hh:mm, "time_zone": str}

[Is static]

ASSOCIATIONS

ATTRIBUTES

 **coordinates** : Private
 Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

Coordinate history array; each entry is lon, lat[, elevation].

[Is static

 **methods** : Private
 Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False)

use <http://codes.wmo.int/wmdr/GeopositioningMethod>

[Is static

 **type** : str Private = MovingPoint

[Is static

 **validFrom** : Private
 Multiplicity: ([1..*], Allow duplicates: 0, Is ordered: False)

Date array aligned with coordinates.

[Is static

ASSOCIATIONS

 Association (direction: Unspecified)

Source: Public facility (Class) Facility
 Cardinality: [1]

Target: Public
 temporalGeom
 Cardinality

ATTRIBUTES	
 methods : Private Multiplicity: ([0..*], Allow duplicates: 0, Is ordered: False) use http://codes.wmo.int/wmdr/GeopositioningMethod [Is static	

ASSOCIATIONS	
 Association (direction: Unspecified) 3-07 temporalGeometry	
Source: Public (Class) ObservationSeries Cardinality: [1]	Target: Public Cardinality
describes	has_geolocatin

Territory

Class in package 'WMDR2'

WMDS reference: 3-02
mandatory for fixed facilities, optional for mobile facilities.
Implemented as [{"date": date, "territory": str}, {}] like other historical<Name> cl

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ATTRIBUTES	
 territory : Private Alias: temporal Territory Values from http://codes.wmo.int/wmdr/TerritoryName [Is static	